

Securing Keyless Entry Systems with GPS Proximity Verification

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Abstract—Modern Passive Keyless Entry (PKE) systems in vehicles are increasingly vulnerable to sophisticated relay attacks, such as roll-jam and rollback attacks, which exploit the wireless communication between key fobs and vehicles. These attacks pose a significant challenge to car manufacturers, as they can bypass traditional encryption and rolling-code technologies. In this paper, we propose an advanced key fob security system that integrates GPS-based proximity detection and AES-128 encryption to counteract such attacks. The proposed system ensures that vehicles respond only to key fobs physically located within a predefined range, verified through encrypted GPS coordinates. Additionally, the system maintains a log of recent GPS locations to detect anomalies indicative of relay attacks or tampering. We used simulations conducted in Proteus and Arduino IDE to demonstrate the effectiveness of this approach in enhancing vehicle security by preventing unauthorized access and detecting potential relay attack attempts. This multi-layered defense not only mitigates existing vulnerabilities but also provides a scalable foundation for future advancements in keyless entry systems.

Index Terms—GPS, Key fob, Atmel ATA5795, MCU (Microcontroller), Relay attacks, Keyless Entry.

I. INTRODUCTION

A. Background and Literature Review

Modern Passive Keyless Entry (PKE) and Remote Keyless Entry (RKE) systems have become more and more popular in recent years, but along with it come serious security issues in the automotive sector. [5] For easy locking, unlocking, and engine starting, these systems rely on wireless communication between a key fob and the car, which allows easy access and a passive experience for car owners. PKE systems also calculate proximity using low-frequency (LF) signals, whereas RKE systems commonly employ ultra-high frequency (UHF) signals for authentication.

However, keyless entry systems are now a primary target for cyber attacks due to their weaknesses. In PKE systems, the vehicle automatically detects the key fob's presence nearby, allowing the doors to unlock without any user action, while in RKE, the user must press a button on the key fob to lock or

unlock the vehicle.. Relay attacks such as rollback and roll-jam attacks are some of the most common types of attacks that car jammers exploit. Relay attacks use key fob signals that have been intercepted to enter cars without authorization. At the same time, roll-jam entails jamming signals and intercepting rolling codes sent between the vehicle and the key fob. Research shows that these attacks are becoming more common across various car brands and models, indicating that the security measures in place are insufficient to counteract sophisticated threats.

One can easily navigate and find countless videos demonstrating replay, roll-jam, and rollback attacks using tools such as Flipper Zero, HackRF One, and Evil Crow RF, to mention a few. These devices are able to intercept, record, and mimic the signal of a vehicle's key fob. In addition to breaking into a car, these devices are able to record and emulate a garage door signal. This would give a thief access not only to a person's vehicle but also to their home.

Our research is based on the review of 'Key Fob Replay Attacks on Personal Vehicles: Vulnerabilities and Mitigation Strategies' research paper [2].

This paper mentioned explores the details of modern key fobs, focusing on their underlying technology, security features, and communication interfaces. It also examines the diverse communication links utilized by these key fobs, the implementation of rolling codes, and code-hopping technology. Then this paper dives into exploiting vulnerabilities within modern key fob systems and ways of mitigation. Mitigation strategies addressed in this paper included an encryption-reinforced timestamp method implemented to protect rolling-code RKE systems from jamming and replay attacks. This approach involves the encrypted signal transmitted by the key fob containing both a timestamp and a counter value. The signal is decrypted upon reception, and the timestamp is cross-checked with the synchronized counter to validate it.

The third paper explored security vulnerabilities in Radio Frequency Identification (RFID) based RKE systems in

modern vehicles. In this paper, the researcher demonstrated radio-based cyber-physical attacks like RollBack and RollJam in order to uncover weaknesses and provide insights into improving these systems against potential exploits. [4]

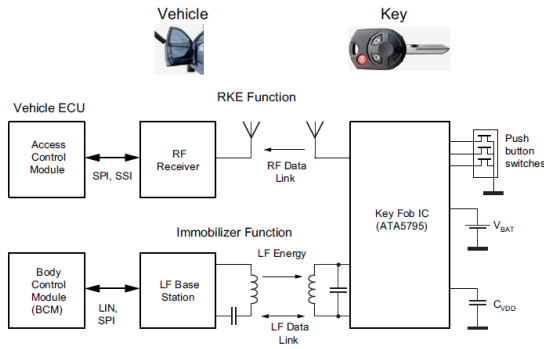


Fig. 1. In-System Integration of Key Fob with GPS for Enhanced Vehicle Security and Communication. [1]

Upon further investigation, we found that rolling codes, encryption, and distance bounding protocols can be defeated with the right equipment. Relay attacks operate at the link layer, making it possible to forward encrypted link layer Protocol Data Units (PDUs). It can also detect encrypted changes to connection parameters (such as connection interval, WinOffset, PHY mode, and channel map) and continue relaying connections through parameter changes. Thus, neither link layer encryption nor encrypted connection parameter changes are defenses against this type of relay attack.

GPS system integration into the key fob for proximity verification and location validation.

II. SYSTEM MODEL

can replay the two unlock signals consecutively, granting access multiple times. This works by causing the vehicle to resynchronize with a previous code when using the first jammed code to roll back, and then the second unlock signal lets the attacker in.

Rolljam attacks similar to Rollback works by jamming the original signal sent to the vehicle and copying it at the same time. Now, when the user sends the second unlock signal, the attacker captures it and jams it again but plays the first unlock signal that was captured. Essentially, the attacker will be able to play the second unlock signal that was captured later if no other unlock signal was used afterward, therefore demonstrating a vulnerability in rolling code-based schemes. [4]

Our approach to mitigate the current system flaws is the integration of GPS technology into key fobs to address the primary limitations of existing security measures. This enhancement will be particularly effective in scenarios where relay attacks undermine traditional encryption. By combining both physical proximity and encrypted GPS verification, the proposed system aligns with emerging trends in vehicle cybersecurity, offering a multilayered approach to securing keyless entry systems, providing the car owner with the utmost security.

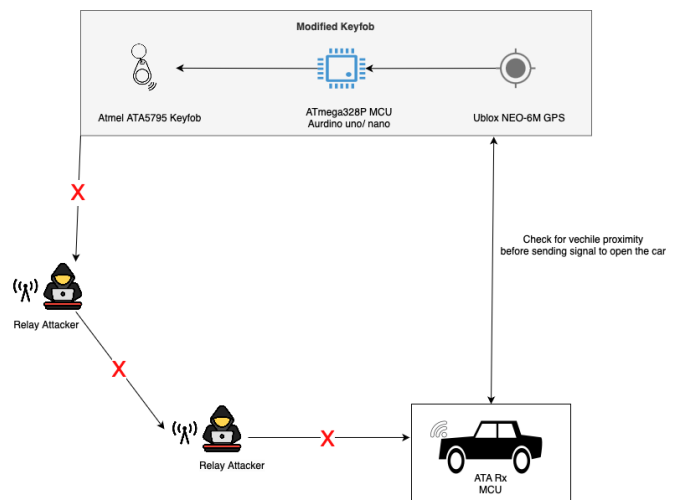


Fig. 2. Workflow of the Proposed Modified Key Fob with GPS for Secure Vehicle Access and Relay Attack Prevention.

A. Microcontroller and GPS Interface:

To integrate GPS functionality into the Atmel ATA5795, we combined the GPS module with the ATA5795, utilizing a Microcontroller (MCU), allowing the system to transmit GPS data over a wireless link. Attackers might attempt to spoof GPS data sent between the key fob and the car, for which we add encryption to the message signal. For our key fob simulation, we chose Atmel ATA5795 key fob, NEO-6M GPS module, and Atmega328 Microcontroller.

Figure 1-2 Microcontroller and GPS interface

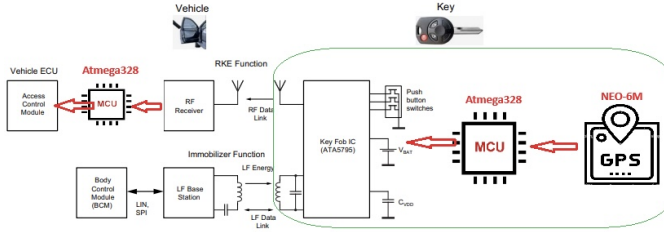


Fig. 3. MCU and GPS Interface for Secure Communication Between the Key Fob and Vehicle ECU.

Atmel ATA5795 is a wireless remote-control device commonly used in automotive and access control systems. It is based on the Atmel ATA5795 SoC, which combines a microcontroller, RF transmitter, and security features like AES encryption to enable low-power, secure wireless communication for locking, unlocking, and other remote functions. [7]

NEO-6M GPS module, a tiny well performing complete GPS receiver with a built-in 25 x 25 x 4mm ceramic antenna, which provides a strong satellite search capability. [8]

Atmega328 as provides various features like GPIO for SPI communication; a UART interface for the NEO6M GPS module: to regulate the data input and transmission of the ATA5795; Depending on the voltage level of the circuit, it can operate on a low power consumption of 3.3V/5V Logic Compatibility and has enough memory to store and process GPS data (such as parsing NMEA phrases). [6]

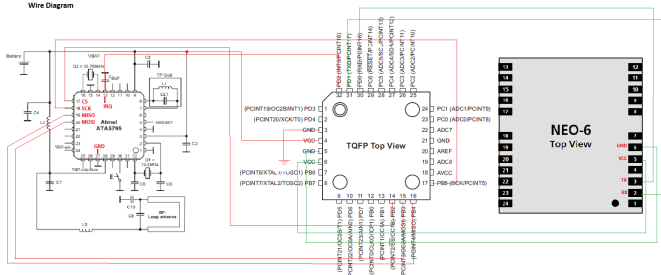


Fig. 4. Wire Diagram Illustrating the Connections Between the Key Fob's MCU and the GPS Module.

Table I outlines the connections between the Atmega328 and ATA5795 IC, detailing the SPI communication pins (MOSI, MISO, SCK, and SS) alongside the power and interrupt

connections (VCC, GND, INT, and IRQ). This configuration enables the Atmega328 to decode and process data from the ATA5795, which handles RF signal demodulation. Table II illustrates the pin mapping between the Atmega328 and the NEO-6M GPS module, focusing on UART communication (TX and RX) and power connections (VCC and GND), facilitating precise location data acquisition for proximity detection. Together, these mappings ensure seamless microcontroller integration with the key fob's essential components, allowing us to add GPS to a key fob.

Atmega328		ATA5795	
Pin 15	MOSI	Pin 20	MOSI
Pin 16	MISO	Pin 19	MISO
Pin 17	SCK	Pin 18	SCK
Pin 14	SS	Pin 17	CS
Pin 4	VCC		Battery
Pin 5	GND	Pin 27	GND
Pin 32	INT	Pin 13	IRQ

TABLE I

PIN MAPPING FOR ATMEGA328 AND ATA5795

Atmega328		NEO-6M	
Pin 5	GND	Pin 6	GND
Pin 6	VCC	Pin 5	VCC
Pin 30	RX	Pin 3	TX
Pin 31	TX	Pin 2	RX

TABLE II

PIN MAPPING FOR ATMEGA328 AND NEO-6M

B. Microcontroller and ECU interface

We must make a few changes in the vehicle's Electronic Control Unit (ECU) to add an ATA5743. This IC demodulates the received RF signal and outputs the raw data on its DATA pin.5 We will add an ATmega328 MCU to the ECU for reading the demodulated data from the ATA5743, decoding it, and processing it (e.g., displaying received messages).

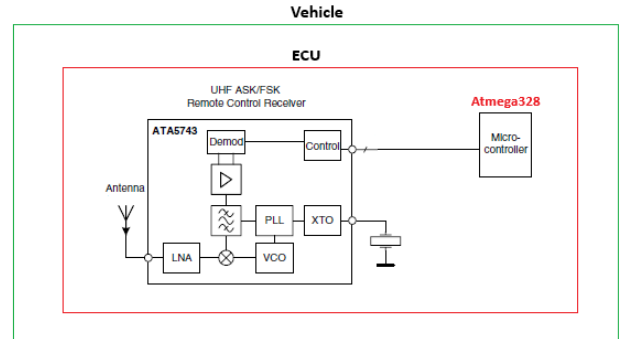


Fig. 5. RX Block Diagram Showing the Receiver Circuit and Integration with the ATmega328 MCU.

Table III highlights the pin mapping between the Atmega328 microcontroller and the ATA5743 IC, which is

responsible for receiving and processing RF signals. Pin 2 of the Atmega328 (digital pin) is linked to Pin 20 of the ATA5743 (Data), which allows the microcontroller to receive decoded RF data. The power and grounding are ensured through connections to VCC (Pin 6 of the Atmega328 and Pin 4 of the ATA5743) and GND/AGND, while the Mode pin (Pin 16 of the ATA5743) configures the IC's operational mode. This setup enables the Atmega328 to process RF signals efficiently for further system operations.

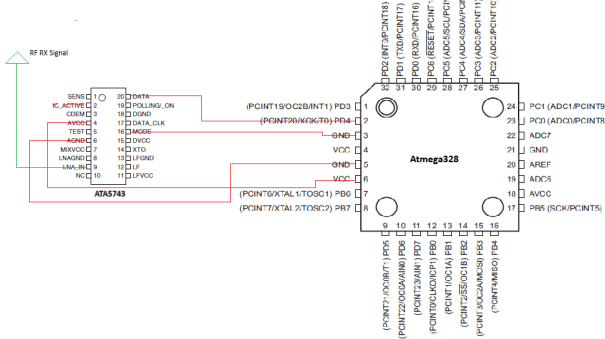


Fig. 6. RX Wiring Diagram Showing the Pin Connections for the ATmega328 MCU and Receiver Module.

Atmega328	ATA5743
Pin 2	Digital pin
Pin 3	GND
Pin 5	GND
Pin 6	VCC
	Pin 4
	AVCC

TABLE III

PIN MAPPING FOR ATMEGA328 AND ATA5743

IV. SIMULATION AND CODING

The proposed system integrates GPS-based proximity detection, AES-128 (Advanced Encryption Standard-128) encryption, and anomaly monitoring to enhance the security of keyless vehicle access systems. The system calculates the distance between the car and the key fob based on the GPS coordinates, allowing the vehicle to unlock only if the key is within a predefined range. For our experiment, we consider the range to be 70 meters, ensuring physical proximity. A log of the key fob's GPS locations is maintained for the past five minutes, every 25 seconds. If we see a location jump while comparing the logs, it suggests relay attack attempts. We use AES-128 encryption to communicate between the key fob and the car, ensuring that only legitimate, verified signals can unlock the vehicle.

A. Car simulation

The car-side simulation in Proteus involves the integration of several components to replicate the behavior of the vehicle's Electronic Control Unit (ECU) in response to signals from the GPS-augmented key fob. We add Atmega328P microcontroller

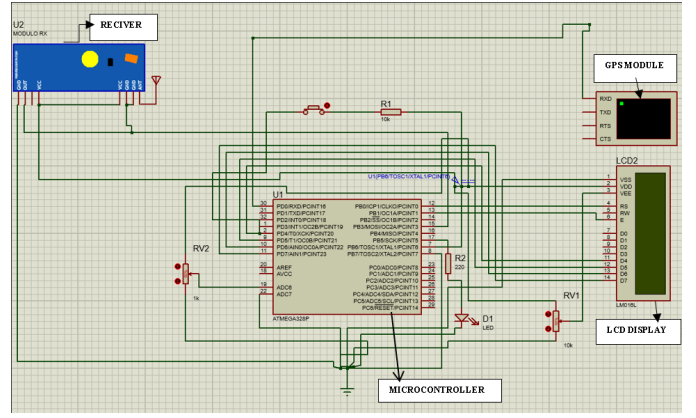


Fig. 7. Simulated Car System in Proteus Demonstrating Component Integration and Vehicle Behavior Replication.

(U1), a GPS module (GPS2), an LCD display (LCD2), and a wireless communication module (U3) to simulate a car in Proteus. The GPS module provides location data to the microcontroller, enabling proximity-based unlocking logic. The LCD display is used to present real-time feedback, such as the distance between the car and the key, or alerts in case of anomalies, like a Relay attack detection or just the GPS location for longitude and latitudes. The wireless module facilitates communication between the car and the key fob. We add a few additional components, such as resistors (R1, R2), potentiometers (RV1, RV2), and an LED (D1) to ensure proper operation. This comprehensive setup successfully simulates how the car verifies encrypted GPS signals from the key and responds to unlock commands when within a predefined range, thereby implementing secure keyless entry.

B. Key fob simulation

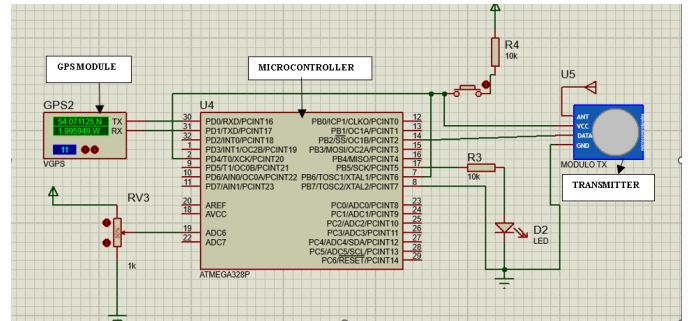


Fig. 8. Simulated Modified Key Fob with GPS in Proteus for Functionality Testing and Integration.

The key fob simulation in Proteus models the functionality of the proposed GPS-augmented keyless entry system. We add the Atmega328P microcontroller (U4) as the central component, which processes GPS data from the GPS module (GPS2) and communicates with the car via the wireless transmitter module (U5). The GPS module provides real-time location coordinates, enabling the key fob to calculate its distance from the vehicle. We add a potentiometer (RV3) for calibration

purposes, while resistors (R3, R4) ensure proper voltage levels for the circuitry. An LED (D2) serves as a visual indicator for operational states, such as successful encryption or error conditions. The microcontroller encrypts the GPS data using AES encryption, ensuring secure communication. The setup is designed to test and validate key fob operations like proximity detection, signal encryption, and communication with the car's receiver module, demonstrating a robust simulation of the enhanced security features for keyless entry systems.

C. Coding

We used Aurdino IDE for software implementation, using C++ and designed a system that does uses GPS-based proximity detection using Haversine Formula and anomaly monitoring using logs to store the GPS co-ordinates. We calculated the distance between the car and the key fob using GPS coordinates using the Haversine formula which is a widely used formula to calculate the shortest distance between tow points in the globe and constitutes to real world. We secured communication by encrypting the unlock command using AES-128 before transmission. We use The Initialization Vector (IV) to ensures that even if the same plaintext and key are used repeatedly, the ciphertext will be different each time. We also maintained a GPS location log every 25 seconds for 5 minutes accounting for 12 log list of the key fob to detect anomalies, such as significant or unexpected location jumps, to flag potential relay attack attempts. The code checks for the key's proximity based on the stored logs. If a jump in the location is found, i.e., when calculating the distance between the logs, if a place is 1000 meters away from the last location, it indicates the signal is coming from an Relay attacker. Also, the car only unlocks if the key is within 70 meters. Along with these features, the message signal communicated between these two entities is encrypted in the keyfob, and the car decrypts it using the same key, which makes the system more secure from attackers.

This paper focuses on areas around the university to conduct the experiment of location jump and detection of really attackers. The car is assumed to be parked with a stationary location (Panther Dining Hall) while the key fob is simulated at various buildings and locations both near and distant from the campus.

V. RESULTS AND DISCUSSIONS

After running the code and simulation the code in Arduino IDE and Proteus based on the algorithm discussed above, we get the following results.

- Proximity-Based Behavior:** The system helps us to determine whether the key is within the defined proximity of the car or not. As shown in figure 10, for locations close to the car (e.g., Evans Library), the car unlocks as expected whereas for a location that is far (e.g., Skurla Hall) it keeps the door closed, ensuring proper functionality of the proximity-based access control.
- Relay attack Detection:** The simulation demonstrates the system's ability to detect suspicious key movements. As

Algorithm 1: Pseudo-Code for GPS Encryption and Anomaly Detection

Data: GPS Coordinates, Encryption Key
Result: Secure communication with anomaly detection

```

1 Function CalculateDistance (lat1, lon1, lat2, lon2):
2   Compute dLat and dLon
3   Use Haversine formula to calculate distance
4   return distance in meters

5 Function EncryptMessage (Message):
6   Encrypt the message using AES-128
7   return encrypted message

8 Function DecryptMessage (EncryptedMessage):
9   Decrypt the encrypted message using AES-128
10  return original message

11 Function DetectRelayattack (LocationLog):
12  foreach entry in LocationLog do
13    Calculate distance between logged coordinates
14    if distance > 1000 meters then
15      return True (Relay attack detected)
16    end
17  end
18  return False

```

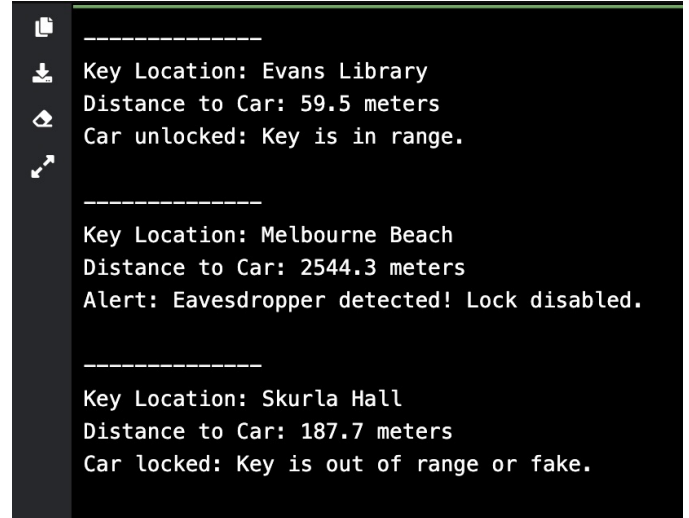


Fig. 9. Output Demonstrating Key Fob Behavior Across Different Locations and Proximity Scenarios.

shown in figure 11, when the cordinates change from Panther Dinning Hall to Melbourne Beach, calculating the Haversine formula reveals the distance difference to be 2500 meters. The sudden change of the location of Key fob suggest a relay attacker sending the signals from a different place. If a jump of location is detected, then the car remains locked, and an alert is sent.

- Boundary Testing:** We also observe far locations such as Skurla Hall, which is moderately far from the car, is

Key Fob GPS Location Log:

Log Index	Name	Latitude	Longitude
0	Panther Dining Hall	28.0658	-80.6244
1	Panther Dining Hall	28.0658	-80.6244
2	Panther Dining Hall	28.0658	-80.6244
3	Panther Dining Hall	28.0658	-80.6244
4	Panther Dining Hall	28.0658	-80.6244
5	Panther Dining Hall	28.0658	-80.6244
6	Panther Dining Hall	28.0658	-80.6244
7	Panther Dining Hall	28.0658	-80.6244
8	Panther Dining Hall	28.0658	-80.6244
9	Melbourne Beach	28.0836	-80.6081
10	Panther Dining Hall	28.0658	-80.6244
11	Panther Dining Hall	28.0658	-80.6244

Haversine Distance : 2548 meters
Alert: Relay Attacker detected! Suspicious location change:
From: Panther Dining Hall (Lat: 28.0658, Lon: -80.6244)
To: Melbourne Beach (Lat: 28.0836, Lon: -80.6081)

Fig. 10. Output Showing Location Jump Detected, Triggering Relay attack Alert in the System.

correctly identified as "out of range." This ensures access is denied to keys not meeting the proximity requirement, validating the system's threshold (70 meters).

```
[NMEA] $GPRMC,075620.814,V,,,,,011224,,N*42
[NMEA] $GPGGA,075620.814,,,0.10,,M,,,*72
[NMEA] $GPGSA,A,3,13,11,20,28,14,18,16,21,22,19,,,,,*3A
[PC=0x0714] [AVR USART 0] RX Frame Error.
[PC=0x0720] [AVR USART 0] RX Frame Error.
[PC=0x0722] [AVR USART 0] RX Frame Error.
[PC=0x2592] [AVR USART 0] RX Frame Error.
[PC=0x0700] [AVR USART 0] RX Frame Error.
[NMEA] $GPRMC,075622.066,V,,,,,011224,,N*4D
[NMEA] $GPGGA,075622.066,,,0.10,,M,,,*7D
[NMEA] $GPGSA,A,3,13,11,20,28,14,18,16,21,22,19,,,,,*3A
[PC=0x2588] [AVR USART 0] RX Frame Error.
[PC=0x06F0] [AVR USART 0] RX Frame Error.
[PC=0x0714] [AVR USART 0] RX Frame Error.
[NMEA] $GPRMC,075623.313,V,,,,,011224,,N*4D
[NMEA] $GPGGA,075623.313,,,0.10,,M,,,*7D
[NMEA] $GPGSA,A,3,13,11,20,28,14,18,16,21,22,19,,,,,*3A
[PC=0x258A] [AVR USART 0] RX Frame Error.
[PC=0x06F0] [AVR USART 0] RX Frame Error.
[PC=0x0704] [AVR USART 0] RX Frame Error.
[NMEA] $GPRMC,075624.566,V,,,,,011224,,N*4E
[NMEA] $GPGGA,075624.566,,,0.10,,M,,,*7E
[NMEA] $GPGSA,A,3,13,11,20,28,14,18,16,21,22,19,,,,,*3A
```

Fig. 11. GPS Coordinates of the Key Fob Location Captured in the Simulated Proteus Environment.

- d) **Accuracy of Distance Calculation:** Using GPS coordinates and the Haversine formula accurately determines the distances between the car and the key. Then the system uses the distance to align with real-world expectations based on the coordinates provided.

The system can effectively prevent unauthorized access to the car by implementing proximity-based unlocking and anomaly detection mechanisms. The encryption of the "open" command adds an additional layer of security to ensure that only authenticated keys can unlock the car.

However, the system depends on pre-defined locations for simulation. Real-world deployment would require integration with a GPS module for live location updates. Detection of Relay attacks relies on analyzing historical logs. The system may raise false positives if the key moves unusually for legitimate reasons.

VI. CONCLUSION AND FUTURE WORK

In this paper, the integration of GPS-based proximity detection with AES-128 encryption provides a significant enhancement to the security and functionality of modern keyless vehicle access systems. By leveraging the calculated GPS coordinates, the proposed system ensures that the vehicle unlocks only when the key fob is within a predefined proximity range, thereby mitigating the risk of unauthorized access. Additionally, the system's ability to monitor and log GPS locations over time allows for detecting abnormal activity, such as relay attacks or spoofing attempts, further enhancing security.

A few enhancements to the system would be adding low-power GPS or activating - deactivating the GPS based on the car's proximity, as the constant use of GPS tracking may cause a problem of draining the keyfob's battery. The project can be tested in diverse scenarios, such as urban areas, rural settings, and underground parking lots.

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